

provements are exciting because they have enabled commercially-viable LED lighting solutions which are now available to the consumer," he adds.

Commenting on other associated developments, Marion Reichl, press officer, OSRAM Opto Semiconductors GmbH, says, "There are several improvements within the last few years that have given LED technology a real drive... ceramic packages for better thermal management; increase in efficiency within a short timeframe, with the help of chip technologies; organic LEDs (OLEDs) as a light source of the future; long-life, high-brightness white LEDs with longer maintenance intervals and better total cost of ownership; and so on. These improvements support the fast and extensive adoption of LEDs for different applications."

"LED is an initiative towards lowering energy demand and consumption, through lower power consumption and higher output," says Hari Chereddi, managing director, Sujana Energy Limited. In this regard, he remarks, "The three most exciting improvements have been in the areas of applicability, availability and affordability of LEDs from the perspective of a common man. Applicability to our day-to-day lighting needs, availability in most shapes and sizes and, most significantly, becoming affordable to all, thereby providing limitless opportunities for 'linking light and life'... Sujana Energy sees this as one of the most exciting areas of growth."

According to Ajay Goel, CEO, Goldwyn Limited, the payback period of LED-based lighting solutions with respect to conventional technologies is as follows:

1. Streetlights for the governments/ municipalities: Immediate as the cost of putting up power plant (for the amount of power saved) will be more than the cost of LED streetlights
 2. Office LED lights: 18 months to 36 months
 3. Other LED lights: 30 months, on an average
- So far so good, and it promises to better, as there is a lot of LED-related

Did we say 'digital'?

You heard it right. An LED is part and parcel of the digital world. In fact, it is a simple, little 'chip'. Even the manufacturing process bears a lot of semblance to that of computer chips. A p-n (positive-negative) junction is created out of a suitable semiconductor material. When electrons flow from the anode to the cathode, they move to a lower energy level, emitting a photon of light. The light could be red, green, blue or amber depending on the materials used to make the diode. The light from different coloured LEDs may be combined or a blue LED lens may be coated with phosphor to make white light.

LEDs are used as components in electrical and electronic devices, as well as for larger applications such as automotive or home lighting.

research happening across the world. "However, major LED manufacturers have the most expertise and also the largest resources to put towards research," comments Thieken.

Not alone, though, the research work is done in partnership, points out Reichl. OSRAM Opto Semiconductors has a broad network of partners in the scientific community with connections at universities and research institutes. The company is also active in projects supported by the public sector (and there is quite a lot of thrust from this front, as LEDs are a good solution to today's power-guzzling lighting equipment).

Evidently, there is a shove from all directions to take LED technology forth. Here is a sample of what is happening and what is in store.

LED in your living room

Efforts are on make LEDs the *de-facto* standard for home and public lighting. Although LED lights are still too costly to motivate the common man to go for them, power savings, long life and low maintenance requirements have resulted in companies opting for LED-based lighting systems for their offices. Today, the most efficient LEDs are about ten times more energy-efficient than incandescent bulbs.

Also, with LEDs, it is possible to develop solidstate lighting (SSL) products, that is, lamps made using LEDs and OLEDs rather than

electrical filaments, plasma or gas. The light weight and no brittle parts make such lamps robust and shock-resistant. The lifespan is also longer. They are now used mostly for traffic lights, remote controls, etc, but soon they might rule the home lighting market.

While it might be a few more years before we replace all the bulbs at home with LED lamps, the process of phasing out the traditional energy-guzzling incandescent bulbs has already begun in several countries, including parts of Australia, Europe and South America. The United States of America is targeting 2012, and is even holding contests such as the L-Prize, to spur lighting manufacturers to develop high-quality, high-efficiency solidstate lighting products. 60W bulbs account for 50 per cent of the domestic incandescent market, and replacing these with LED bulbs

would save enough electricity per year to light 17.4 million households. It would also avoid 5.6 million metric tonnes of carbon emissions annually. The US Department of Energy hopes this to be the outcome of the L-Prize.

A list of the 50 best inventions of 2009 published in *TIME* magazine in November 2009 included Philips' entry to the L-Prize (which also happens to be the first entry to the contest). Philips' LED bulb emits the same amount of light as its incandescent equivalent but uses less than 10 watts and lasts 25 times



Philips' MASTER LED lamp